

AthDGC · Athens-PROIEL

An open diachronic Greek treebank with Indo-European parallels

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The **Athens Diachronic Glossa Chronos** (*Athens-PROIEL*) is an open PROIEL-style dependency treebank spanning Greek diachrony with Indo-European parallels.

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AthDGC is **both** a continuously-updated **PROIEL XML 2.0** treebank **and** an open-source **computational diachronic-linguistics toolkit**:

- LightSIDE-compatible feature extraction + classifier training
- NoSketch-style concordancer
- Fine-tuned Stanza checkpoints (Byzantine, Late-Byzantine, Modern adaptations)
- Argument-structure extractor
- Cross-lingual alignment viewer (Neo4j)
- Quarto template pack for any DH project (this very deck is built with it)

Covers:

- Homeric and Archaic Greek (Homer, Hesiod, Sappho, Pindar)
- Classical (Aeschylus, Sophocles, Euripides, Aristophanes, Plato, Aristotle)
- Koine and NT (Plutarch, Strabo, Lucian, the New Testament)
- Late Antique (Eusebius, Basil, Chrysostom, Procopius)
- Byzantine (Psellos, Anna Komnene, John of Damascus, Niketas Choniates)

with **NT verse-level cross-alignment** to Latin (Vulgate), Gothic (Wulfila), and Old Church Slavonic (Marianus).

Metric	Value
Total corpus rows	89.9 M
Annotated Greek rows	4.08 M
NT-aligned Greek verses	6,861
IE cross-aligned witnesses	4

1. **Discovery** - daily harvest from archive.org, Perseus, First1K, Wikisource, Diorisis
2. **Filtering** - Greek-script ratio, apparatus-criticus rejection, dedup
3. **Conversion** - PROIEL XML 2.0
4. **Annotation** - Stanza `grc_proiel` + analogous models per parallel language
5. **Argument-structure** - subject, object, oblique, voice, aspect per verb
6. **Cross-alignment** - LaBSE + AwesomeAlign (mBERT)
7. **Storage** - PROIEL XML + JSONL + Qdrant + Neo4j

- **Retranslation** of influential texts (Iliad, NT, Septuagint Psalms, classical historiography)
- **Retelling** chains across periods and languages
- **Argument structure** under retranslation

- Lavidas (PI) · Nikiforidou · Haug (Oslo) · Kulikov (Ghent)
- Geka · Symeonidis · Michalareas
- Chionidi · Tsiropina · Plakoutsi · Argyropoulos

NKUA · Athens Digital Glossa Chronos Research Network

Lavidas, N., Nikiforidou, K., Haug, D., Kulikov, L., Geka, V., Symeonidis, V., Michalareas, T., Chionidi, S., Tsiropina, A., Plakoutsi, E., and Argyropoulos, E. (2026). *AthDGC: Athens Diachronic Glossa Chronos*. Zenodo.

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- Showcase: <https://athdgc.github.io>
- Press: https://athdgc.github.io/athdgc_press.html
- GitHub: <https://github.com/AthDGC>
- HFRI project site: <https://athdgc.github.io/hfri2024>

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